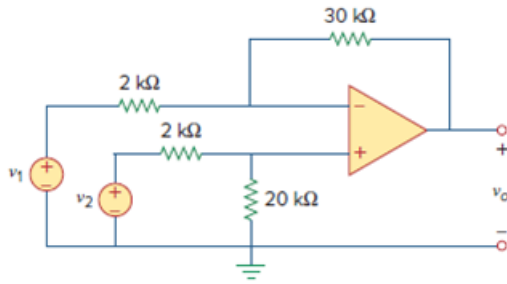


Problem

The circuit in Fig. 5.79 is for a difference amplifier. Find v_o given that $v_1 = 1$ V and $v_2 = 2$ V.

Figure 5.79:



Step-by-step solution

Step 1 of 3

Consider the following difference op-amp circuit with nodes marked as a and b .

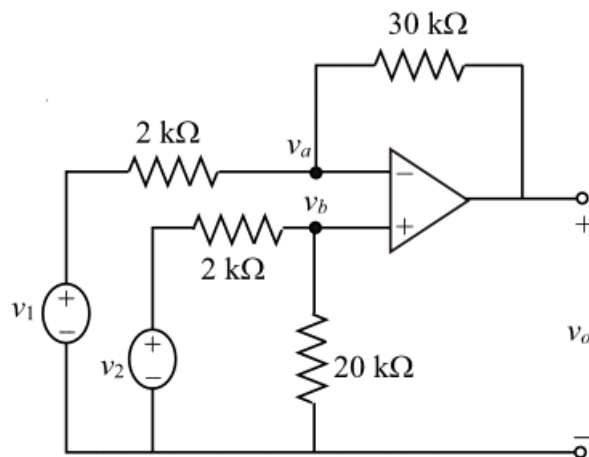


Figure 1

Step 2 of 3

The input voltages are,

$$v_1 = 1 \text{ V}$$

$$v_2 = 2 \text{ V}$$

Apply Kirchhoff's current law at node a .

$$\frac{v_1 - v_a}{2 \text{ k}} = \frac{v_a - v_o}{30 \text{ k}}$$

Substitute 1 for v_1 .

$$\frac{1 - v_a}{2 \text{ k}} = \frac{v_a - v_o}{30 \text{ k}}$$

$$\left(\frac{1}{2 \text{ k}} + \frac{1}{30 \text{ k}} \right) v_a - \frac{1}{30 \text{ k}} v_o = \frac{1}{2 \text{ k}}$$

$$\frac{16}{30} v_a - \frac{1}{30} v_o = \frac{1}{2}$$

$$16v_a - v_o = 15 \dots\dots (1)$$

Apply Kirchhoff's current law at node b .

$$\frac{v_2 - v_b}{2 \text{ k}} = \frac{v_b - 0}{20 \text{ k}}$$

$$\frac{v_2 - v_b}{2} = \frac{v_b}{20}$$

$$v_2 - v_b = \frac{v_b}{10}$$

$$10v_2 - 11v_b = 0$$

Substitute 2 for v_2 .

$$(10)(2) - 11v_b = 0$$

$$11v_b = 20$$

$$v_b = \frac{20}{11}$$

$$v_b = 1.818 \text{ V}$$

But, $v_a = v_b$

Thus, $v_a = v_b = 1.818 \text{ V}$

Substitute 1.818 V for v_a in equation (1).

$$(16)(1.818) - v_o = 15$$

$$v_o = (16)(1.818) - 15$$

$$= 29.088 - 15$$

$$= 14.088 \text{ V}$$

$$\approx 14.09 \text{ V}$$

Thus, the output voltage, v_o of the difference amplifier, is 14.09 V .

[Comments \(1\)](#)



Anonymous

not quite sure how the v_b on the right side disappears in the fourth equation